Thrust to Weight Hand Analysis

- Thrust per motor = 1 kg
- Total thrust = 1 kg × 4 = 4 kg
- Mass of drone components w/o arm = 1 kg
- thrust to weight ratio = 2

Max mass of drone + payload = thrust / thrust to weight ratio

Max mass = 4kg/2 = 2kg

Material	carbon fiber composite	Aluminum alloy	Fiberglass composite	PLA plastic	ABS plastic	wood "birch"
Density (ρ) (kg/m^3)	1600	2700	1900	1250	1050	600

Volumes from CAD software

$V_{armA} = 1.2891 \times 10^{-5} m^3$ $V_{armB} = 1.3211 \times 10^{-5} m^3$ $V_{armC} = 2.8466 \times 10^{-5} m^3$

Mass of 4 arms = $(V \times \rho) \times 4$

Material	"CFRP"	Aluminum Alloy	"GFRP"	PLA Plastic	ABS Plastic	wood
m arms A (kg)	0.0825024	0.1392228	0.0979716	0.064455	0.0541422	0.0309384
m arm B (kg)	0.0845504	0.1426788	0.1004036	0.066055	0.0554862	0.0317064
m arm C (kg)	0.1821824	0.3074328	0.2163416	0.14233	0.1195572	0.0683184

Payload capacity = max mass − ($m_{drone\ components} - m_{drone\ arms}$)

Material	"CFRP"	Aluminum Alloy	"GFRP"	PLA Plastic	ABS Plastic	Wood
arm A (kg)	0.9174976	0.8607772	0.9020284	0.935545	0.9458578	0.969062
arm B (kg)	0.9154496	0.8573212	0.8995964	0.933945	0.9445138	0.9682936
arm C (kg	0.8178176	0.6925672	0.7836584	0.85767	0.8804428	0.9316816